

Internet of Things (IoT)

Lecture 1: Introduction

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References

- Hanes, D., Salgueiro, G., Grossetete, P., Barton, R., & Henry, J. (2017). IoT fundamentals: Networking technologies, protocols, and use cases for the internet of things. Cisco Press.
https://ptgmedia.pearsoncmg.com/images/9781587144561/samplepages/9781587144561_CH08.pdf
- Iqbal, M. A., Hussain, S., Xing, H., & Imran, M. A. (2020). Enabling the Internet of Things: Fundamentals, Design and Applications. John Wiley & Sons.
<https://onlinelibrary.wiley.com/doi/epub/10.1002/9781119701460>
- Buyya, R., & Dastjerdi, A. V. (Eds.). (2016). Internet of Things: Principles and paradigms. Elsevier.
https://dphoto.lecturer.pens.ac.id/lecture_notes/internet_of_things/Internet%20of%20Things%20Principles%20and%20Paradigms.pdf

LMS



Course Materials: Lectures and Lab Assignments



Online Quizzes



Project Submission



Lab Assignment Submission

Grades



Midterm: 15 Degree



Activities 60

Project: 20 Degree

Quizzes (Practical/LMS): 20

**Practical Assignments: 20
Degree (LMS + Discussion)**



Final Exam: 25 Degree

Course Learning Outcomes

- CLO 1:** Define and explain the concept of the Internet of Things (IoT), including its key components, enabling technologies, and significance in modern systems.
- CLO 2:** Describe the evolution and historical development of IoT, identifying major milestones, contributors, and transitions from traditional networks.
- CLO 3:** Identify and categorize major application areas of IoT, such as smart homes, healthcare, industry, agriculture, and libraries.
- CLO 4:** Analyze the benefits and challenges of implementing IoT systems, including issues related to scalability, security, data analytics, and device interoperability.

IoT Course Topics

- 1. Introduction to IoT**
- 2. IoT Architecture & Network Components**
- 3. Smart Objects & IoT Devices**
- 4. Communication Technologies in IoT**
- 5. IoT Protocol Stacks**
- 6. IoT Application Protocols & Data Formats**
- 7. IoT Security Principles**
- 8. Cloud & Fog Computing in IoT**
- 9. Data Analytics in IoT**
- 10. IoT Platforms & Middleware**
- 11. Designing IoT Solutions**



Introduction to IoT



Agenda

- 1: Understanding IoT
- 2: Evolution & History
- 3: Importance & Scope
- 4: Enabling Technologies
- 5: Benefits of IoT
- 6: IoT Applications
- 7: Challenges in IoT
- 8: Summary & Discussion



Think !!

What is IoT?



What is IoT?

- ✓ Goal of IoT is to “connect the unconnected”
 - “Things” or “objects” that were not supposed to be connected to the Internet



- IoT did the **technology transition** in traditional computer networks

Cont...

- **IoT is the Unifications of technologies:**

- ✓ Embedded Systems
- ✓ Low Power and Low Rate Network
- ✓ Internet
- ✓ Cloud Computing
- ✓ Data Analytics
- ✓ Bigdata
- ✓ Edge Intelligence
- ✓ Network Security and Data Security
- ✓ Software Defined Networks
- ✓ Etc.

- *Alternate Definition:*

“The Internet of Things (IoT) is the **network of physical objects** that contain embedded technology to **communicate** and **sense** or **interact** with their **internal states** or the **external environment**.” – Gartner Research*

Different Names of IoT

Internet of Everything

Smarter Planet

Machine to Machine (M2M)

The Fog

Tsensors (Trillion Sensors)

The Industrial Internet

Industry 4.0

Internet of Things (IoT)

Brief History of Initial Phase of IoT

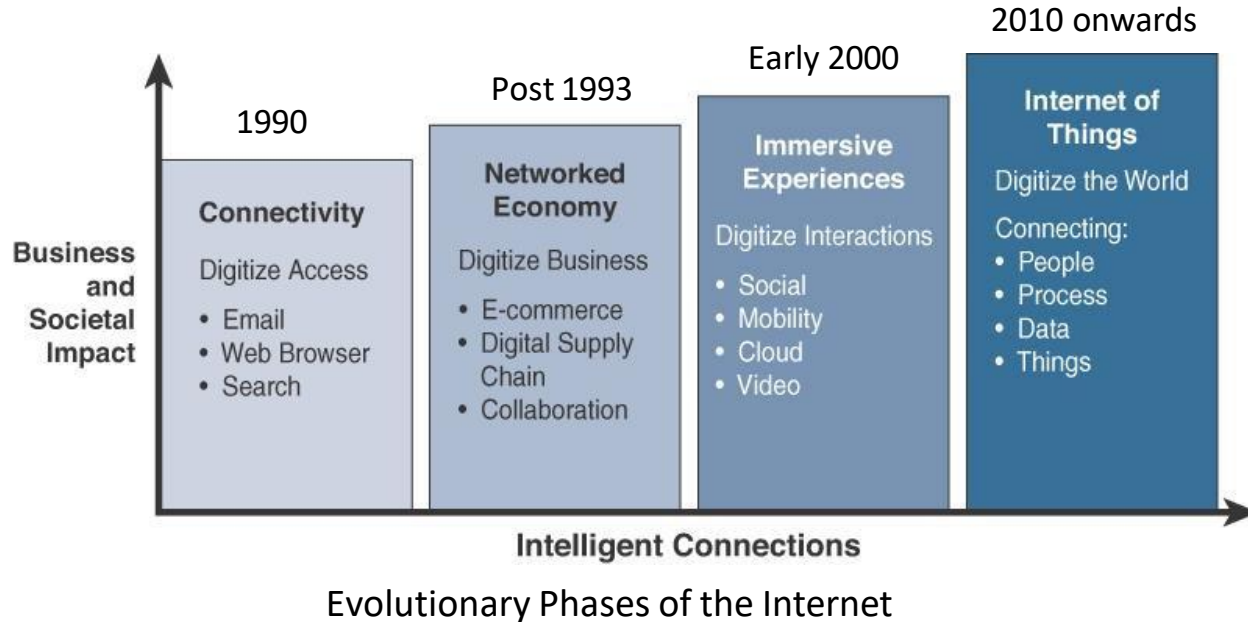
- The term "**Internet of Things**" was likely coined by **Kevin Ashton** of Procter & Gamble, later MIT's Auto-ID Center, **in 1999**.
- Kevin Ashton said –
 - "In 20th century, **computers were brains without senses** —they only knew what we told them."
 - "Now in 21st century, **computers are sensing things for themselves!**"
- **Early 1980s** at the Carnegie Mellon University, a group of students created a way to get their campus **Coca-Cola vending machine** to report on its contents **through a network** in order to save them the trek if the machine was out of Coke.

Brief History of Initial Phase of IoT

- **In 1990**, John Romkey, developer of the first TCP/IP stack for IBM PC in 1983, connected a toaster to the internet for the first time.
- **In 1991**, a group of students at the University of Cambridge used a web camera to report on coffee available in their computer labs coffee pot.
- At the **beginning of the 21st Century**, LG Electronics introduced the world's first refrigerator connected to the internet

Brief History of Initial Phase of IoT

- The popularity of IoT did not accelerate until 2010/2011 and reached mass market from 2013-14.
- Definition of the IoT has also evolved over time.



Benefits of IoT

- **Real-Time Monitoring**
 - IoT sensors can **monitor equipment and processes** in real-time, providing valuable sensed data that can be used to streamline operations, reduce waste, and increase output.
- **Automation of Processes**
 - **Machines can assemble parts** with more precision and speed, resulting in fewer errors during assembly
 - **Robots can** very rapidly **detect faults** that may not be detected by the human eye
- **Improved Efficiency or Productivity**
 - IoT helps to **enhance productivity by** streamlining processes, automating tasks, and providing real-time data insights.
- **Predictive Maintenance**
 - Continuous monitoring of systems and processes to **identify key indicators of problems** before they result in downtime or system failure

Benefits of IoT

- **Improved or New Insights**

- IoT generates vast amounts of data that **can be analyzed to identify trends, inefficiencies, and areas for improvement**. Organizations can leverage these insights to enhance productivity and operational effectiveness.

- **Cost Reduction**

- When an **organization can improve** system uptime, automate processes, reduce the risk of failure and gain insights that support better decision making, and reduce resource usage, the result is efficiency and cost savings

- **Optimized Work Environments**

- IoT technology can help **create more comfortable and efficient workspaces**

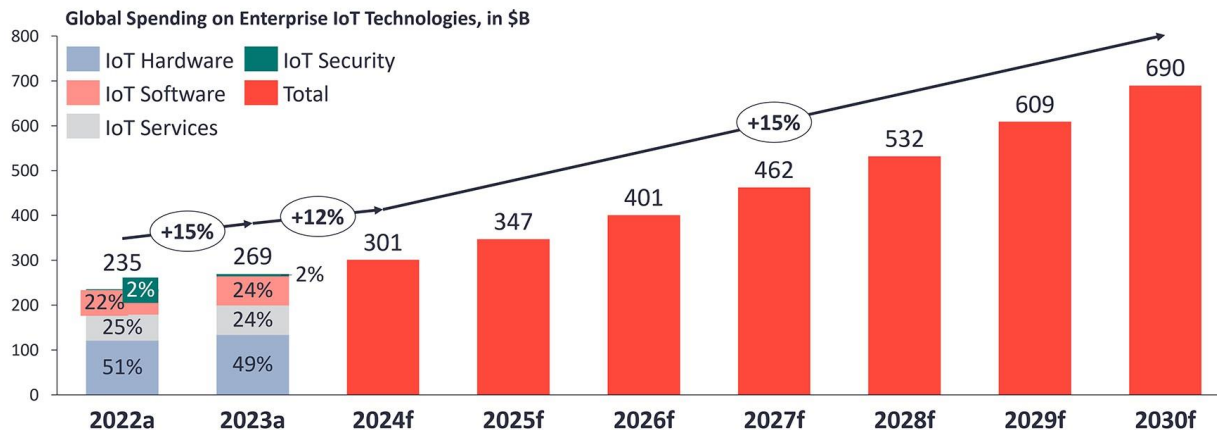
- **Adaptability**

- The **ability to adapt** to new business requirements, customer needs, and changing conditions, or scale the deployment in response to business growth or customer requirements

Global Spending on IoT

The enterprise IoT market by technology 2023–2030

Data as of June 2024



Note: IoT Analytics defines IoT as a network of internet-enabled physical objects. Objects that become internet-enabled (IoT devices) typically interact via embedded systems, some form of network communication, or a combination of edge and cloud computing. The data from IoT-connected devices is often used to create novel end-user applications. Connected personal computers, tablets, and smartphones are not considered IoT, although these may be part of the solution setup. Devices connected via extremely simple connectivity methods, such as radio frequency identification or quick response codes, are not considered IoT devices. Since the last update in 2023 our definition of the enterprise IoT tech stack slightly changed.
a: Actuals, f: Forecast
Source: IoT Analytics Research 2024 – Global IoT Enterprise Spending Dashboard (Q2/2024 update). We welcome republishing of images but ask for source citation with a link to the original post or company website.

IoT Analytics' Report on Global IoT Market as on July 2024

Image source: <https://iotbusinessnews.com/2024/07/10/06205-enterprise-iot-market-size-reached-269-billion-in-2023-with-growth-deceleration-in-2024/>

Where is IoT?

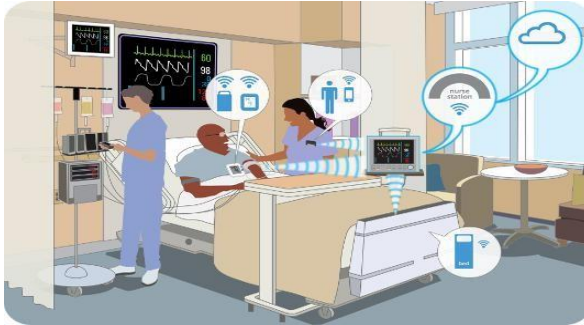


Wearable
Tech Devices



Smart Appliances

It's everywhere!

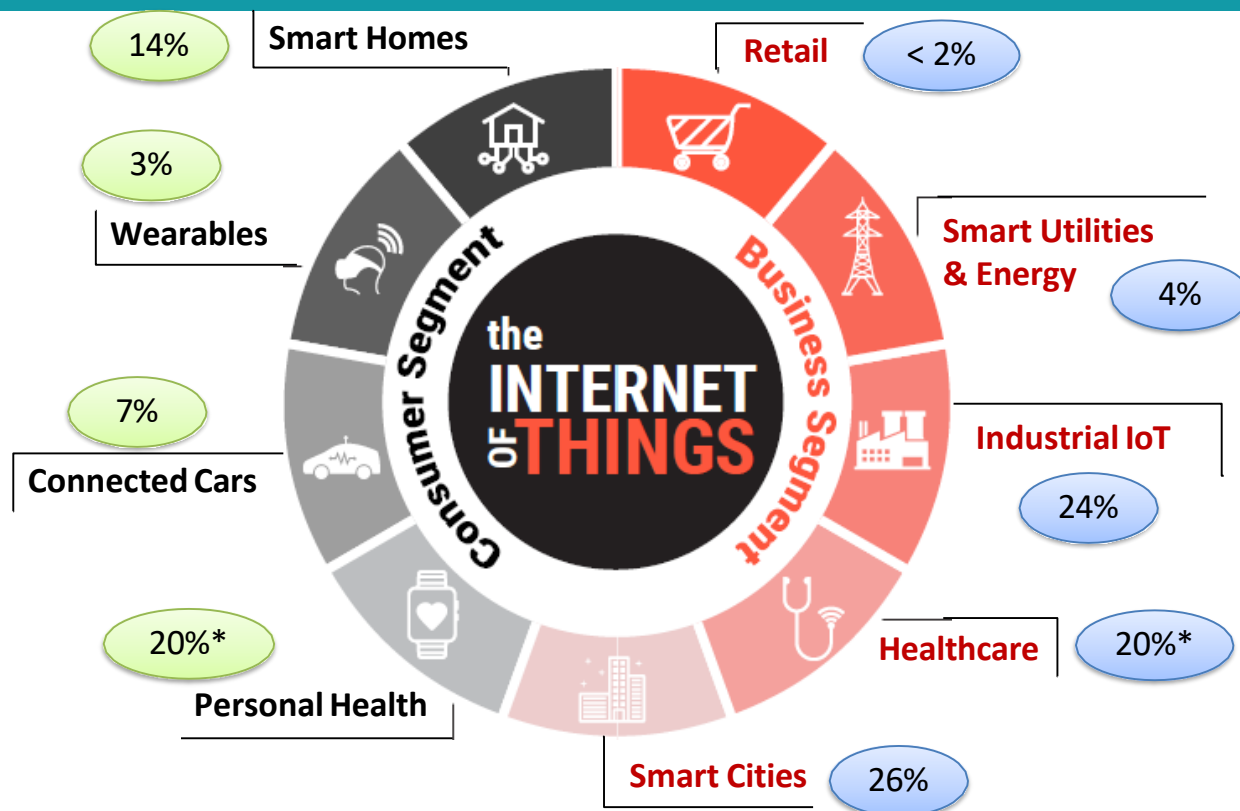


Healthcare



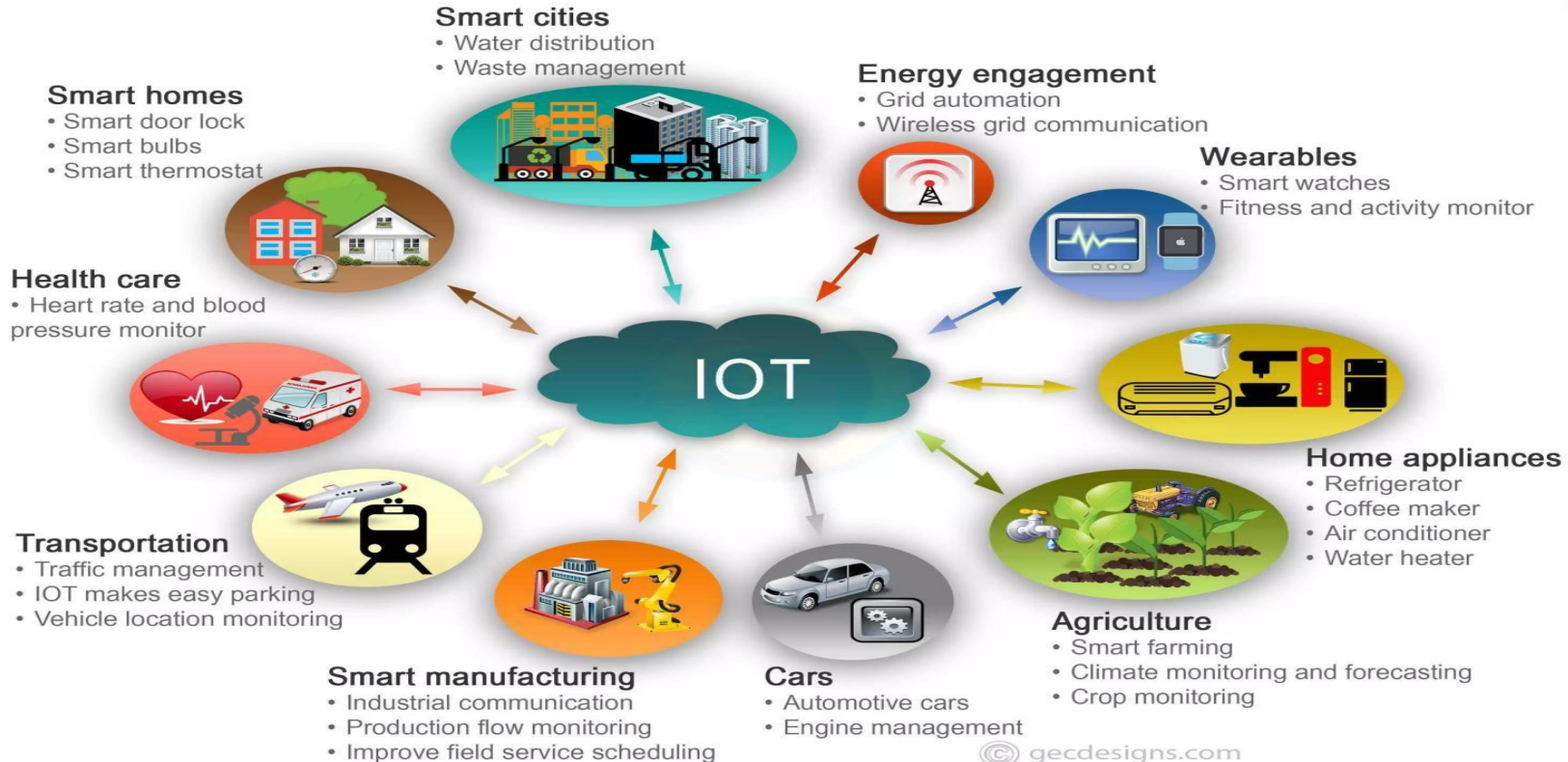
Industry Automation
and Monitoring

Global IoT Market Share



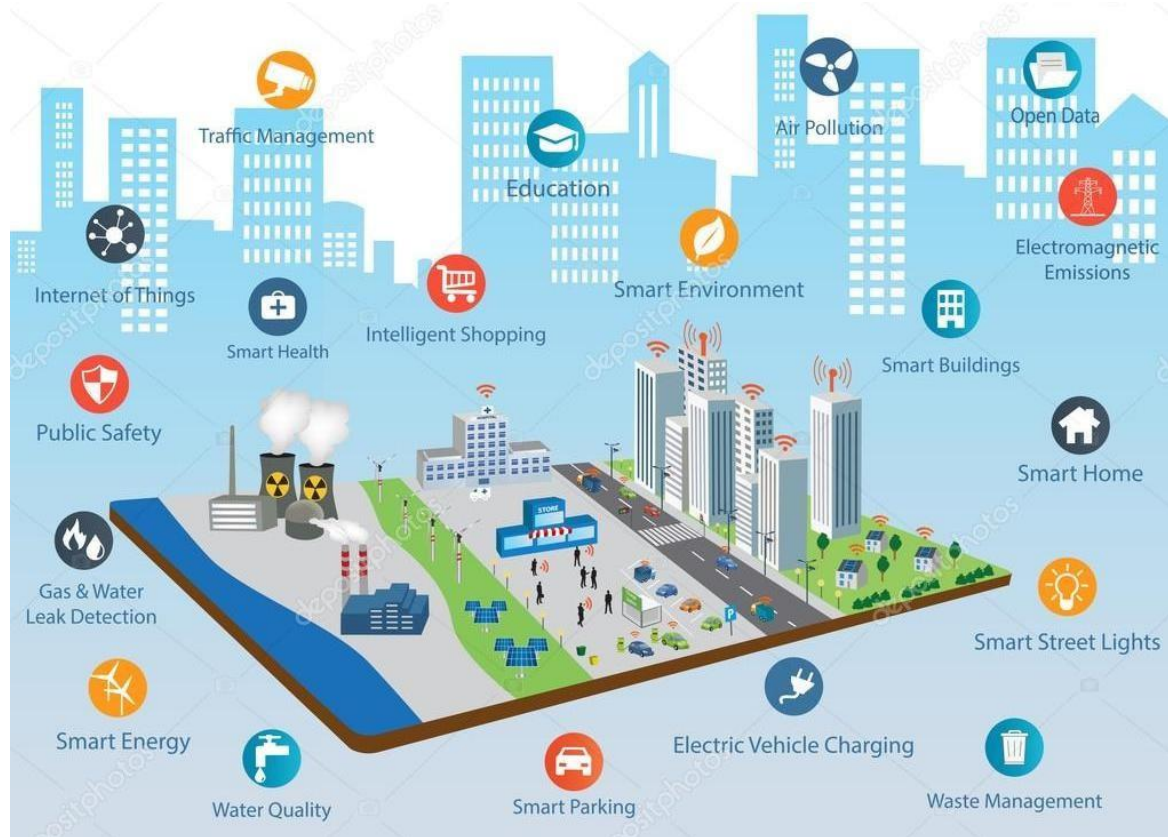
Source: <https://growthenabler.com/flipbook/pdf/IOT%20Report.pdf>

Applications of IoT



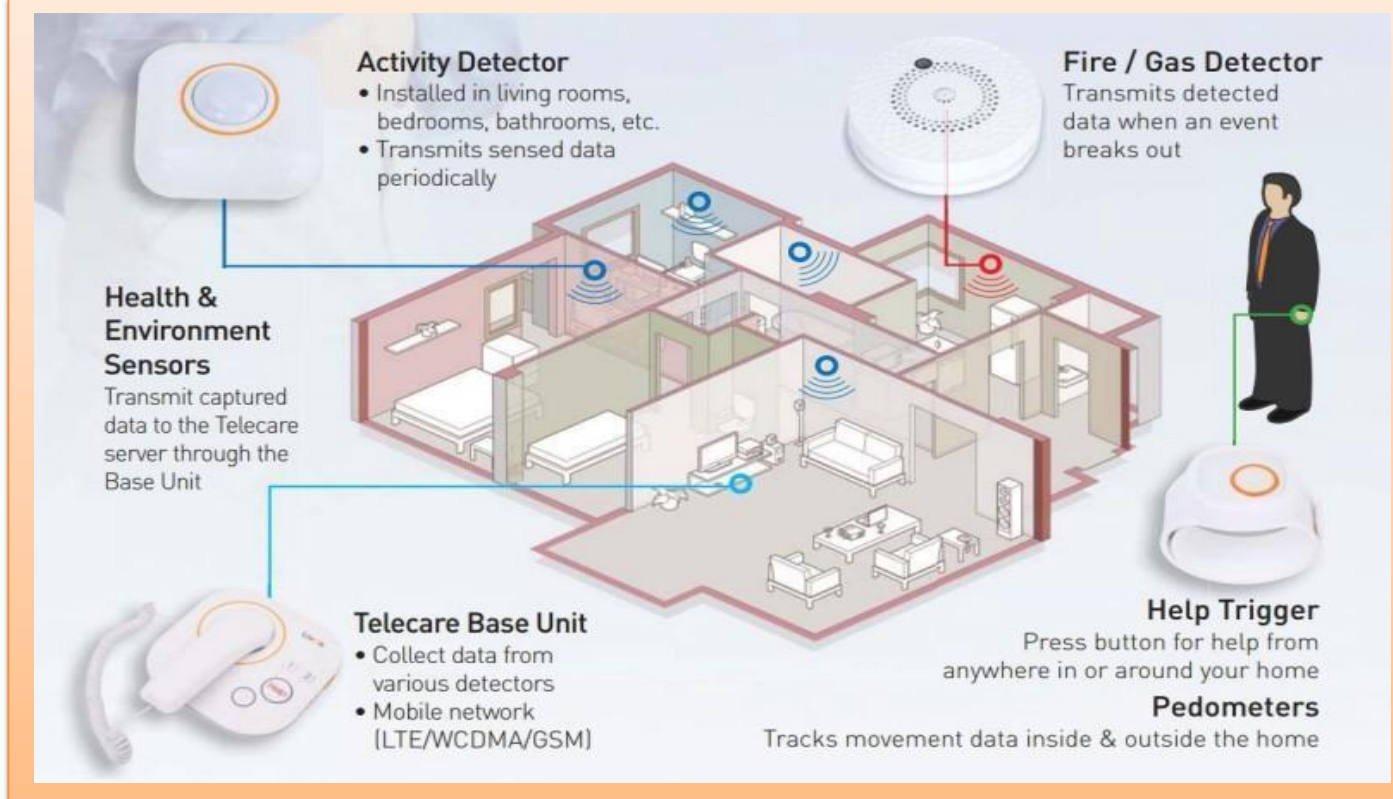
Smart City

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Source: <https://depositphotos.com/126025652/stock-illustration-smart-city-concept-and-internet.html>

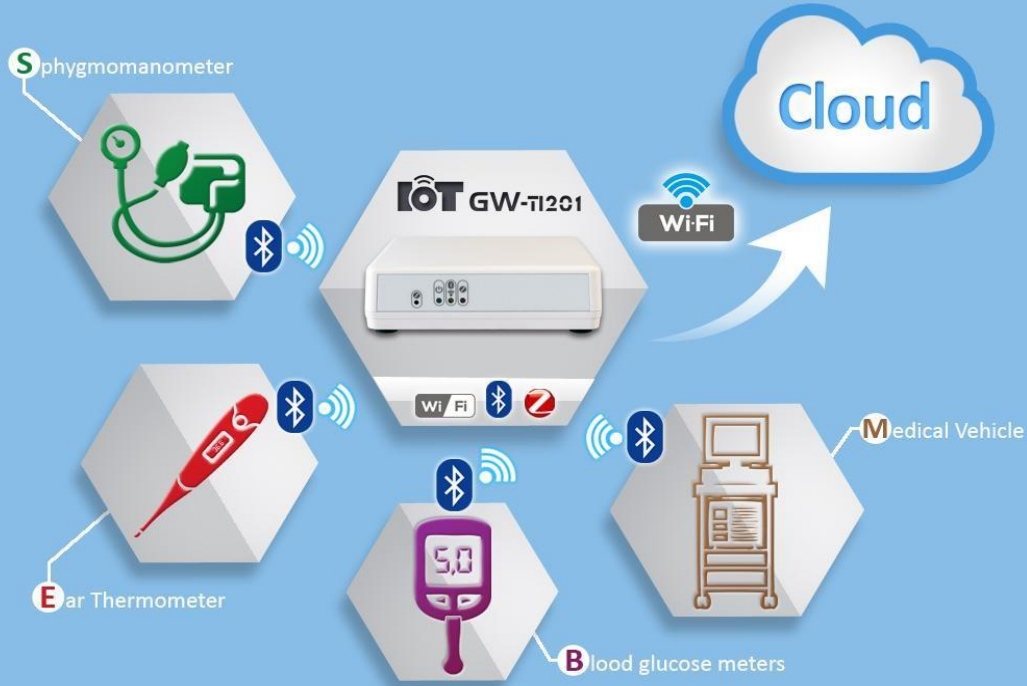
Smart Home



Source: <https://medium.com/@globalindnews/north-america-accounted-for-major-share-in-the-global-smart-home-healthcare-market-in-2015-cc9cc1974ac5>

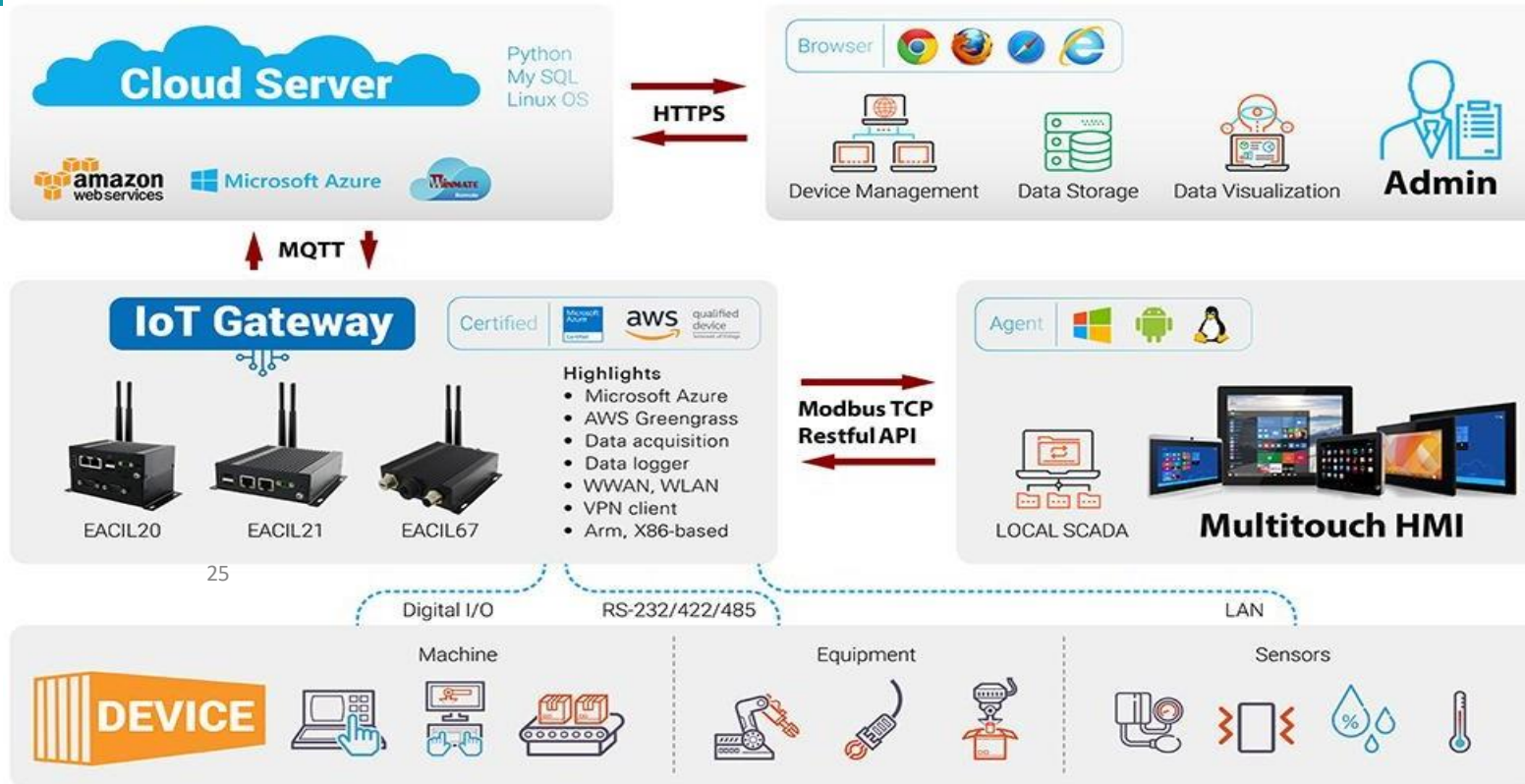
Smart Healthcare

Smart Healthcare - Medical Application

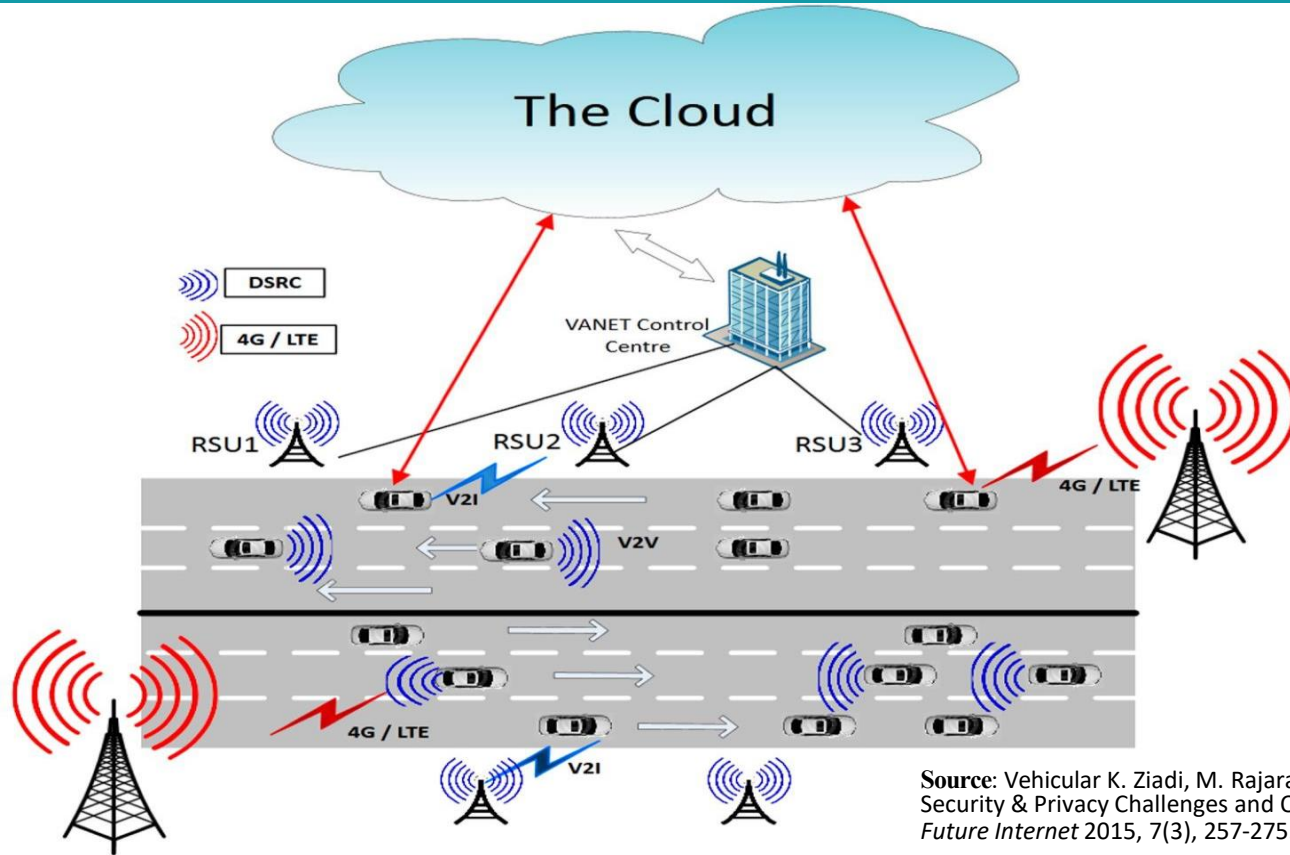


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Industrial IoT



Connected Cars



Source: Vehicular K. Ziadi, M. Rajarajan, "Internet: Security & Privacy Challenges and Opportunities", *Future Internet* 2015, 7(3), 257-275.

Google's Self-Driving Car



Source: <https://www.google.com/>

Smart Agriculture



Source: <https://in.pinterest.com/pin/515380751093603767/?lp=true>

Main Challenges in IoT Implementation

Sensors

- limited resources
- limited types of sensors

Scale

- millions of devices are connected to form IoT

Low Power Network

- devices should remain connected to the network for years
- high network latency
- can't use traditional communication protocols

Interoperability

- various protocol, various architecture
- unavailability of standardized platform
- different technology leads to interoperability issue

Bigdata & Data analytics

- massive amount of sensor data
- different sources and various forms
- extract intelligence form the heaps of data

Privacy

- which personal data to share with whom
- how to control

Security

- “things” becomes connected, so security becomes complex

Lessons Learned

- Learned about what is IoT
- Learned the genesis of IoT
- Understand the benefits of IoT
- Learned about the market share of IoT
- Understand the real world applications of IoT
- Understand various challenges in IoT implementation



Think Questions

How does the Internet of Things (IoT) represent a technological transition from traditional computer networks?

(Hint: Consider how devices interact, process, and respond without direct human input.)

Traditional networks primarily connected computers and relied heavily on human input for data exchange. In contrast, IoT enables physical devices (**sensors, actuators, appliances**) to automatically collect, transmit, and respond to data without human intervention. IoT expands networking to include “things” that were never intended to be connected before, like refrigerators, cars, or wearables, using embedded systems, wireless technologies, and cloud computing. It marks a shift from human-centric to machine-driven communication.



Think Questions

Imagine you are building a smart home system. What "things" would you connect, and what data would you collect? How would this data improve the user's experience?

A smart home could include connected lights, thermostat, security cameras, door locks, smoke detectors, and appliances. Sensors might collect data such as temperature, motion, humidity, air quality, or energy usage.

This data improves user experience by:

1. Automatically adjusting lighting or heating.
2. Sending alerts during security breaches
3. Turning off unused appliances to save energy
4. Enhancing comfort and safety.

-It also enables remote control via smartphones and improves efficiency through automation and analytics.



Think Questions

Among the challenges in IoT implementation (security, privacy, interoperability, power), which do you think is the **most critical** to solve today—and why?

Security is arguably the most critical challenge. As IoT devices scale up and handle sensitive data (e.g., in healthcare or smart homes), they become targets for cyber-attacks. Most IoT devices have limited processing power, making encryption and authentication difficult.

Vulnerabilities like default passwords, outdated firmware, or unsecured interfaces expose users to threats like data theft, surveillance, or DDoS attacks. Thus, securing the IoT ecosystem is vital to its safe and sustainable adoption.





Question
&
Answer

